From Shelf Order to Later Agent Choice: A Link-by-Link Audit of Recommendation Writeback

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Abstract

When an AI shopping agent selects a product, that choice may shape what the platform recommends next. Yet the choice itself may already reflect how the platform arranged the products. We ask whether this influence remains visible after the selected item enters account history. In a research storefront with 460 constructed tasks, two sequential recommenders, and two LLM providers, we trace this process one link at a time. Reordering identical candidate records changed immediate choices. Writing those selections into history then produced small, measurable differences in later shelves and later agent choices for both providers. An extension using ordinary-language shopping tasks also found order-sensitive choices. These findings motivate separating task-local assistance from decisions about what enters account history. The study follows one step of model–recommender feedback; whether it amplifies over repeated interactions remains unresolved.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Human-centered computing** → *Interactive systems and tools.*

Keywords

human-agent interaction, delegated agents, recommender systems, recommendation order, writeback

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1 Introduction

A shopping agent sees an ordered shelf. Once stored in account history, its choice may alter later shelves. LLM order effects and agentic commerce are established [1, 2, 17]. ABxLab tests attribute and nudge effects in agent shopping [4]; recommender work traces position-biased feedback into item popularity [5]. Agent4Rec, AFL, RecInter, and RecLoop further model feedback between simulated agents and recommenders [3, 7, 14, 16]. These systems study interaction over time. We focus on a specific transition: whether an order-only difference remains visible after the selected item is written back and a learned recommender produces the next shelf.

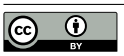
Our experiment separates the initial order effect from its consequence for later input. First, candidate identities and attributes stay fixed while order changes. Next, the selected item enters history as a standardized event; the recommender generates a later shelf for another categorical choice. Selected-item identity can now differ between arms: this second contrast follows a mediated path rather than another order-only intervention (Figure 1).

2 Research Storefront and Protocols

We built 460 tasks, 115 per category, from four public Amazon Reviews 2018 sequences [9]. Study 1 used frozen SASRec checkpoints [8]. Studies 2 and 3 crossed every task with SASRec and GRU4Rec [6], producing 920 task-model units. During model calls, the LLMs received pseudonymous IDs and derived scalar features instead of titles, histories, or review text. DeepSeek V4 Flash and MiMo V2.5 ran at temperature zero in August 2026. The frozen protocol assigned DeepSeek the primary contrast and MiMo a second-provider replication across available API systems. Supplementary Material S1 includes complete cached prompt examples, provider-specific wrappers, output constraints, and a paired-input audit; the computational artifact retains protocols, code, and result primitives.

Study 1: activation. We replayed the same behavior and candidate masks while changing whether temporary task events were logged or activated in the learned recommender. This comparison isolates recommender activation from event recording.

Study 2: order to action. For every unit and provider, we used four input constructions: rank metadata only, inspection state only, a byte-identical masked control, and served order. The served-order comparison kept candidate IDs and attributes fixed and changed only their order. Within each provider, system instructions, task text, serialization, output constraints, and decoding settings stayed fixed across arms. Each arm received two calls, yielding 29,440 valid responses. We used the categorical energy construction [10]: twice cross-arm disagreement minus the two within-arm disagreements. Under stable, independent, exchangeable calls, its mean task-conditional U-statistic estimates squared L_2 distance between categorical output distributions. Two-call unit estimates are noisy and may be negative. We formed intervals by resampling tasks within category while retaining both recommender families and every mode. We froze these rules and the controls before running the study.



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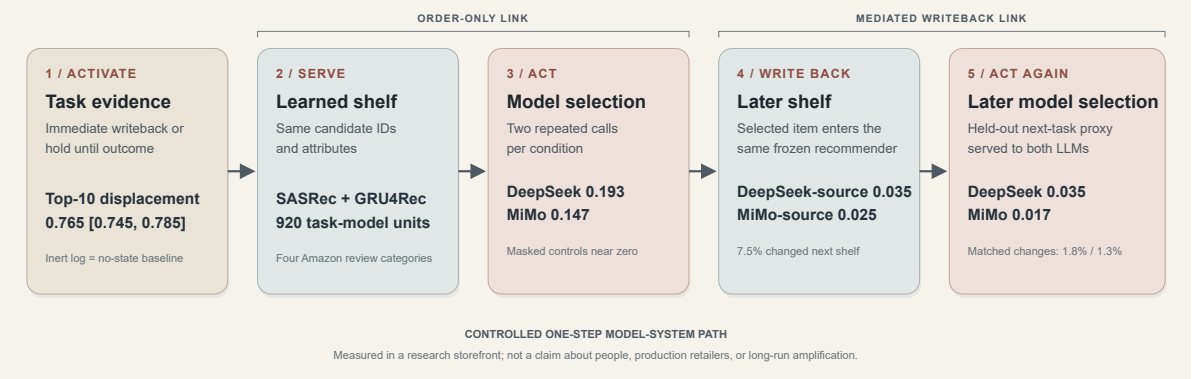


Figure 1: Measured path from shelf order to later agent choice. The order-only link fixes candidate identity and attributes; the mediated writeback link follows the selected item through one later shelf and choice.

Study 3: writeback. We appended each Study 2 selection to the same frozen recommender history as a standardized positive event. We then added one held-out next event and generated a new Top-10. The source provider received this later shelf with a held-out next-task proxy. Each provider retained all 920 units. Within category, recommender, and mode strata, a frozen four-row cyclic Latin square assigned the four semantic conditions to call positions; counts differed by at most one. The same 226 protocol-hash-selected units received a byte-identical downstream control.

Product-language extension. In 64 tasks crossed with both recommenders (128 units; 60 with a feasible offered item), both providers saw ordinary-language requests and the same 12 cleaned title, brand, and price records in arm-specific order. We scored task outcomes privately and continued each trajectory for three fixed steps. The continuation compared selected events with one deterministic random realization matched on write count and whether paired writes were equal.

3 Results

Logging without activation did nothing. Immediate activation versus holding produced a mean in-task Top-10 displacement of 0.765 [0.745, 0.785], while no-state and inert-log shelves agreed. At resolution and the standardized next task, fixed replay returned to zero displacement.

Order-only exposure changed categorical choices for both providers. DeepSeek's distance was 0.193 [0.151, 0.239], compared with 0.004 [-0.003, 0.013] in the masked control. MiMo's distance was 0.147 [0.110, 0.187], versus 0.000 [-0.007, 0.008] in its control. All cross-arm changes were swaps between selectable candidates, not select-to-abandon transitions.

The selected-item difference remained measurable after writeback. DeepSeek-sourced selections produced later-shelf distance 0.035 [0.024, 0.047]; MiMo-sourced selections produced 0.025 [0.016, 0.035]. Estimates were positive for SASRec (0.018 [0.010, 0.027]) and GRU4Rec (0.052 [0.034, 0.071]). Serving later shelves back to their source provider produced distances 0.035 [0.017, 0.054] for DeepSeek and 0.017 [0.005, 0.032] for MiMo; both 226-unit byte-identical controls were zero. Matched repeat-pair change rates were 1.8% [1.0%, 2.8%] (34/1,840 comparisons; 18/920 units with any change) and 1.3% [0.7%, 2.0%] (24/1,840; 15/920 units). MiMo's SASRec-only estimate was 0.007 [-0.004, 0.020], while GRU4Rec was 0.028 [0.009, 0.052].

Across the 128 product-language units, DeepSeek and MiMo order distances were 0.309 [0.180, 0.453] and 0.289 [0.180, 0.414]. Success differences (holding minus immediate) were 0.031 [-0.008, 0.078] and 0.023 [-0.016, 0.066]; regret and failure also included zero. At step three, the selected-versus-random difference was 0.007 [-0.023, 0.035] for DeepSeek. For MiMo it was -0.017 [-0.036, -0.001]. Point estimates did not consistently favor selected writeback, but one random realization cannot establish equivalence or absence of amplification.

4 Implications and Limits

The chain identifies two design decisions: how the platform presents candidates, and when an agent's selection becomes account history. Prior systems separate intent-specific profiles [11], recommendation memories [15], memory admission [12], or agents from platforms [13]. Our study follows an order-induced choice through one writeback cycle. This motivates an interface that separates task-local assistance from admission to account history, with task outcome informing that admission. Evaluating that design requires a direct policy comparison and measures of the principal's goals.

The evidence concerns model–recommender behavior, with no human participants. Tasks and next-task targets are constructed proxies; token-and-price success is not human utility. Later pilot steps use a fixed policy rather than repeated LLM calls. Two available providers are not representative of LLMs, and two repeats per arm limit stationarity assessment. DeepSeek returned one provider fingerprint; MiMo returned none. The one-step horizon and 1.3–1.8% matched endpoint rates establish neither human harm nor long-run amplification.

Generative AI disclosure. Generative AI tools assisted with code generation, experiment orchestration, and language editing. The author checked the protocols, statistical decisions, sources, and reported results against the frozen artifacts included with the submission.

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